

BGP

B O R D E R G A T E W A Y P R O T O C O L

C O M P R E H E N S I V E L A B M A N U A L — R e v
3

eBGP Configuration & Verification

Best Path Selection • Neighbor Relationships • IPv4 & IPv6 • Authentication

Topology	AS 100 / AS 147 / AS 180
Difficulty	Intermediate–Advanced
Duration	5–7 Hours
Protocols	eBGP, iBGP, OSPF, MD5 Auth
Address Families	IPv4 Unicast, IPv6 Unicast

1. Lab Topology Overview

This lab uses a three-AS topology. All routers run Cisco IOS-XE. OSPF is the IGP within each AS. Study the addressing and OSPF domain boundaries carefully before starting.

1.1 Autonomous Systems Summary

AS	Router(s)	Interfaces	Role	IGP
AS 100	Apolo, Beta	Apolo: Gi0/0(→Charlie), Gi0/1(→Beta) Beta: Gi0/0(→Apolo), Gi0/1(→Echo)	Transit / iBGP Hub	OSPF Area 0
AS 147	Delta, Echo	Delta: Gi0/0(→Echo) Echo: Gi0/0(→Delta), Gi0/1(→Beta)	eBGP Multihop peer / Non-BGP transit	OSPF Area 0
AS 180	Charlie	Gi0/0(→Apolo)	Single-homed eBGP customer	OSPF Area 0

1.2 IP Addressing Table

Router	Interface	Connected To	IPv4 Network	IPv4 Address	IPv6 Address
Charlie	Gi0/0	Apolo Gi0/0	102.10.20.0/24	102.10.20.1/24	2001:db8:1::1/64
Charlie	Lo0	Router-ID only	1.1.1.1/32	1.1.1.1/32	—
Charlie	Lo1	Advertised prefix	192.168.180.0/24	192.168.180.1/24	2001:db8:180::1/64
Apolo	Gi0/0	Charlie Gi0/0	102.10.20.0/24	102.10.20.2/24	2001:db8:1::2/64
Apolo	Gi0/1	Beta Gi0/0	102.16.11.0/24	102.16.11.1/24	2001:db8:a::1/64
Apolo	Lo0	Router-ID only	2.2.2.2/32	2.2.2.2/32	—
Apolo	Lo1	Advertised prefix	10.100.0.0/24	10.100.0.1/24	2001:db8:100:1::1/64
Beta	Gi0/0	Apolo Gi0/1	102.16.11.0/24	102.16.11.2/24	2001:db8:a::2/64
Beta	Gi0/1	Echo Gi0/1	172.16.18.0/24	172.16.18.1/24	2001:db8:3::1/64
Beta	Lo0	Router-ID + multihop	4.4.4.4/32	4.4.4.4/32	2001:db8:100::2/128
Beta	Lo1	Advertised prefix	10.100.1.0/24	10.100.1.1/24	2001:db8:100:2::1/64
Echo	Gi0/1	Beta Gi0/1	172.16.18.0/24	172.16.18.2/24	2001:db8:3::2/64
Echo	Gi0/0	Delta Gi0/0	192.168.55.0/24	192.168.55.2/24	2001:db8:2::2/64
Delta	Gi0/0	Echo Gi0/0	192.168.55.0/24	192.168.55.1/24	2001:db8:2::1/64

Router	Interface	Connected To	IPv4 Network	IPv4 Address	IPv6 Address
Delta	Lo0	Router-ID + multihop	3.3.3.3/32	3.3.3.3/32	2001:db8:147::1/128
Delta	Lo1	Advertised prefix	172.20.0.0/24	172.20.0.1/24	2001:db8:147:1::1/64

NOTE	Lo0 on each router is the BGP router-ID and (for Delta and Beta) the multihop update-source. Lo1 on each router is a dedicated prefix injected into BGP via the 'network' statement — giving a realistic prefix to advertise without using infrastructure addresses.
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1.3 OSPF Domain Boundaries

OSPF Domain	Routers	Networks Included
AS 100 — Area 0	Apolo, Beta	102.10.20.0/24, 102.16.11.0/24, 172.16.18.0/24, 2.2.2.2/32, 4.4.4.4/32, 10.100.0.0/24, 10.100.1.0/24
AS 147 — Area 0	Delta, Echo	192.168.55.0/24, 172.16.18.0/24, 3.3.3.3/32, 172.20.0.0/24

KEY	172.16.18.0/24 (Beta Gi0/1 ↔ Echo Gi0/1) is an inter-AS link included in BOTH AS 100 OSPF and AS 147 OSPF — exactly the same pattern as 102.10.20.0/24 (Charlie↔Apolo) which is in AS 100 OSPF. This allows loopback reachability without static routes.
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1.4 BGP Session Map

Session	Peers	Type	Notes
Charlie Gi0/0 ↔ Apolo Gi0/0	AS 180 ↔ AS 100	eBGP direct	Single hop, TTL=1
Delta Lo0 ↔ Beta Lo0	AS 147 ↔ AS 100	eBGP Multihop	2 hops via Echo; TTL=2
Apolo Gi0/1 ↔ Beta Gi0/0	AS 100 ↔ AS 100	iBGP	102.16.11.0/24 link
Beta Gi0/1 ↔ Echo Gi0/1	AS 100 ↔ AS 147	No BGP	OSPF transit for multihop TCP
Echo Gi0/0 ↔ Delta Gi0/0	AS 147 internal	No BGP	OSPF within AS 147

2. Lab Objectives

Upon completing this lab you will be able to:

1. Configure and verify eBGP between Charlie (AS 180) and Apolo (AS 100) — directly connected on Gi0/0.
2. Configure OSPF within AS 100 and AS 147 including the 172.16.18.0/24 inter-AS transit link.
3. Configure and verify eBGP Multihop between Delta (AS 147) and Beta (AS 100) using Lo0 loopbacks with ebgp-multihop 2.
4. Configure iBGP between Apolo and Beta within AS 100 with next-hop-self.
5. Apply best-path selection using weight, LOCAL_PREF, MED, and AS-path prepending.
6. Enable MP-BGP IPv6 using IOS-XE address-family ipv6 unicast with all neighbor declarations inside the address-family block.
7. Configure and verify BGP MD5 authentication on all sessions.
8. Verify IPv4 and IPv6 BGP tables and end-to-end reachability across all three ASes.

3. Pre-Lab: BGP Concepts Review

3.1 BGP Terminology

Term	Definition
eBGP	BGP session between routers in different ASes. Default TTL=1 (direct connection).
eBGP Multihop	eBGP across multiple hops; TTL raised with ebgp-multihop N to exact hop count.
iBGP	BGP within same AS. Next-hop preserved; next-hop-self overrides this.
update-source	Forces BGP TCP session to source from a specified interface (loopback for stability).
next-hop-self	iBGP speaker replaces eBGP next-hop with its own address so iBGP peers can resolve it.
MD5 authentication	Shared password hashed into BGP TCP segments; both peers must match.
network statement	Injects a prefix into BGP only if an exact match exists in the routing table.

3.2 BGP Best-Path Selection Algorithm

Applies in order — first criterion that produces a winner ends the process:

9. Highest Weight (Cisco proprietary, local to router only).
10. Highest LOCAL_PREF (iBGP scope — propagates to all iBGP peers).
11. Locally originated (network/redistribute/aggregate preferred over learned).
12. Shortest AS_PATH length.
13. Lowest ORIGIN code (IGP < EGP < Incomplete).
14. Lowest MED (compared only within same AS unless always-compare-med enabled).
15. eBGP preferred over iBGP.
16. Lowest IGP metric to BGP next-hop.
17. Oldest eBGP path.
18. Lowest BGP Router-ID.
19. Shortest cluster-list (Route Reflector environments).
20. Lowest neighbour IP address.

3.3 Multihop Reachability via OSPF and OSPFv3

The Delta↔Beta eBGP Multihop session (Lo0 3.3.3.3 ↔ Lo0 4.4.4.4 for IPv4; 2001:db8:147::1 ↔ 2001:db8:100::2 for IPv6) travels through Echo as transit — exactly 2 hops. Reachability of each loopback is provided by OSPF (IPv4) and OSPFv3 (IPv6), not static routes:

- AS 147 OSPF includes Delta Lo0 (3.3.3.3/32), 192.168.55.0/24, AND 172.16.18.0/24 (Echo Gi0/1). This means Delta learns the path to Beta's side via Echo.

- AS 100 OSPF includes Beta Lo0 (4.4.4.4/32), 172.16.18.0/24 (Beta Gi0/1), AND 102.16.11.0/24. This means Apolo and Beta know how to reach Echo's side.
- 172.16.18.0/24 sits between two ASes — both OSPF domains include it on their respective routers, the same way 102.10.20.0/24 is included in AS 100 OSPF even though it connects to Charlie (AS 180).

KEY

ebgp-multihop 2 is correct: Delta Gi0/0 → Echo → Beta Gi0/1 = 2 hops. Setting a higher value than necessary is a security risk as it allows BGP packets from unexpected sources to reach the router.

3.4 OSPFv3 IOS-XE Syntax

OSPFv3 on IOS-XE uses 'router ospfv3 PROCESS-ID' (not 'ipv6 router ospf'). Address families are declared inside the process block. Interfaces are enrolled per-interface using 'ospfv3 PROCESS-ID ipv6 area AREA' (not 'ipv6 ospf PROCESS area'). Passive interfaces are set inside the address-family block:

- Process: router ospfv3 1 → address-family ipv6 unicast → passive-interface loopback 0
- Interface: interface Gi0/0 → ospfv3 1 ipv6 area 0
- Verify: show ospfv3 neighbor | show ospfv3 1 ipv6 | show ipv6 route ospf

3.5 IPv6 BGP Neighbor Declaration (IOS-XE)

On IOS-XE, IPv6 BGP neighbors are declared AND activated entirely within the address-family ipv6 unicast block. There is no need to declare them under 'router bgp' first (which would trigger a peer-group error). All neighbor commands — remote-as, description, ebgp-multihop, update-source, next-hop-self — go inside the address-family block.

4. Part 1: OSPF Pre-Configuration

Configure OSPF on all routers before starting BGP. BGP requires IGP reachability for next-hop resolution, loopback advertisement, and multihop transit.

4.1 AS 100 OSPF (Apolo + Beta)

```
! — Apolo —————
Apolo(config)# router ospf 1
Apolo(config-router)# router-id 2.2.2.2
Apolo(config-router)# network 2.2.2.2 0.0.0.0 area 0      ! Lo0 (router-id)
Apolo(config-router)# network 10.100.0.0 0.0.0.255 area 0 ! Lo1 (BGP prefix)
Apolo(config-router)# network 102.10.20.0 0.0.0.255 area 0 ! Gi0/0 to Charlie
Apolo(config-router)# network 102.16.11.0 0.0.0.255 area 0 ! Gi0/1 to Beta
Apolo(config-router)# passive-interface loopback 0
Apolo(config-router)# passive-interface loopback 1

! — Beta —————
Beta(config)# router ospf 1
Beta(config-router)# router-id 4.4.4.4
Beta(config-router)# network 4.4.4.4 0.0.0.0 area 0      ! Lo0 (router-id +
multihop source)
Beta(config-router)# network 10.100.1.0 0.0.0.255 area 0 ! Lo1 (BGP prefix)
Beta(config-router)# network 102.16.11.0 0.0.0.255 area 0 ! Gi0/0 to Apolo
Beta(config-router)# network 172.16.18.0 0.0.0.255 area 0 ! Gi0/1 to Echo
(transit)
Beta(config-router)# passive-interface loopback 0
Beta(config-router)# passive-interface loopback 1
```

4.2 AS 147 OSPF (Delta + Echo)

```
! — Delta —————
Delta(config)# router ospf 1
Delta(config-router)# router-id 3.3.3.3
Delta(config-router)# network 3.3.3.3 0.0.0.0 area 0      ! Lo0 (router-id +
multihop source)
Delta(config-router)# network 172.20.0.0 0.0.0.255 area 0 ! Lo1 (BGP prefix)
Delta(config-router)# network 192.168.55.0 0.0.0.255 area 0 ! Gi0/0 to Echo
Delta(config-router)# passive-interface loopback 0
Delta(config-router)# passive-interface loopback 1

! — Echo — participates in BOTH OSPF domains —————
! AS 147 OSPF process
Echo(config)# router ospf 1
Echo(config-router)# router-id 5.5.5.5
Echo(config-router)# network 192.168.55.0 0.0.0.255 area 0 ! Gi0/0 to Delta
Echo(config-router)# network 172.16.18.0 0.0.0.255 area 0 ! Gi0/1 to Beta

! AS 100 OSPF process (separate process ID on Echo)
Echo(config)# router ospf 2
Echo(config-router)# router-id 5.5.5.5
Echo(config-router)# network 172.16.18.0 0.0.0.255 area 0 ! Gi0/1 shared with
AS100
```

NOTE

Echo runs two OSPF processes: process 1 for AS 147 (with Delta) and process 2 for AS 100 (with Beta) — both include the 172.16.18.0/24 link. This is what allows Delta's Lo0 (3.3.3.3) to be learned by Beta and vice versa, without any static routes.

4.3 Verify OSPF Before Starting BGP

```
! On Beta — should see Delta's loopback via OSPF
Beta# show ip ospf neighbor
Beta# show ip route ospf
! Expected: O 3.3.3.3/32 [110/x] via 172.16.18.2, GigabitEthernet0/1
! Expected: O 172.20.0.0/24 [110/x] via 172.16.18.2

! On Delta — should see Beta's loopback via OSPF
Delta# show ip route ospf
! Expected: O 4.4.4.4/32 [110/x] via 192.168.55.2, GigabitEthernet0/0
! Expected: O 10.100.1.0/24 [110/x] via 192.168.55.2

! Verify multihop path reachability before BGP
Delta# ping 4.4.4.4 source 3.3.3.3
Beta# ping 3.3.3.3 source 4.4.4.4
! Both must succeed before proceeding to BGP configuration
```


5. Part 2: Configure eBGP and iBGP IPv4

5.1 Task 1 – eBGP Direct: Charlie (AS 180) Gi0/0 ↔ Apollo (AS 100) Gi0/0

Charlie and Apollo are directly connected on 102.10.20.0/24. Charlie uses Lo1 (192.168.180.0/24) as its advertised prefix. This must exist in Charlie's routing table before the network statement will work.

Step 1 – Configure Loopbacks on Charlie

```
! — Charlie (AS 180) —————
Charlie(config)# interface loopback 0
Charlie(config-if)# ip address 1.1.1.1 255.255.255.255
Charlie(config-if)# exit
Charlie(config)# interface loopback 1
Charlie(config-if)# ip address 192.168.180.1 255.255.255.0
Charlie(config-if)# no shutdown
Charlie(config-if)# exit
```

Step 2 – Configure eBGP

```
! — Charlie (AS 180) —————
Charlie(config)# router bgp 180
Charlie(config-router)# bgp router-id 1.1.1.1
Charlie(config-router)# no auto-summary
Charlie(config-router)# no synchronization
Charlie(config-router)# neighbor 102.10.20.2 remote-as 100
Charlie(config-router)# neighbor 102.10.20.2 description eBGP-to-Apollo-Gi0/0
Charlie(config-router)# network 192.168.180.0 mask 255.255.255.0
Charlie(config-router)# end

! — Apollo (AS 100) —————
Apollo(config)# interface loopback 0
Apollo(config-if)# ip address 2.2.2.2 255.255.255.255
Apollo(config)# interface loopback 1
Apollo(config-if)# ip address 10.100.0.1 255.255.255.0
Apollo(config-if)# exit

Apollo(config)# router bgp 100
Apollo(config-router)# bgp router-id 2.2.2.2
Apollo(config-router)# no auto-summary
Apollo(config-router)# no synchronization
Apollo(config-router)# neighbor 102.10.20.1 remote-as 180
Apollo(config-router)# neighbor 102.10.20.1 description eBGP-to-Charlie-Gi0/0
Apollo(config-router)# network 10.100.0.0 mask 255.255.255.0
Apollo(config-router)# end
```

Step 3 – Verify

```
Apollo# show bgp ipv4 unicast summary
Neighbor      V  AS      MsgRcvd  MsgSent  TblVer  InQ   OutQ   Up/Down
State/PfxRcd
102.10.20.1   4   180      15       12       5       0     0     00:08:32  1
```

```
! Verify Charlie's Lo1 is in Apolo's BGP table and RIB
Apolo# show bgp ipv4 unicast 192.168.180.0/24
! Look for '>': best path, next-hop 102.10.20.1

! Ping Charlie Lo1 from Apolo
Apolo# ping 192.168.180.1 source 102.10.20.2
! Expected: !!!!! – Lo1 is up on Charlie and installed in Apolo's RIB
```

WARN

If 'show bgp ipv4 unicast 192.168.180.0' shows the prefix without '>' it is NOT in the RIB. Most common cause: 192.168.180.0/24 does not exist in Charlie's routing table — verify 'show ip route 192.168.180.0' on Charlie shows a connected route via Lo1.

5.2 Task 2 – eBGP Multihop: Delta (AS 147) Lo0 ↔ Beta (AS 100) Lo0

With OSPF already advertising Delta Lo0 (3.3.3.3) into AS 147 and Beta Lo0 (4.4.4.4) into AS 100 — and both reachable via Echo — we can now configure the eBGP Multihop session. The path is exactly 2 hops: Delta Gi0/0 → Echo Gi0/0 → Echo Gi0/1 → Beta Gi0/1.

Step 1 – Configure Loopbacks on Delta and Beta

```
! — Delta (AS 147) —————
Delta(config)# interface loopback 0
Delta(config-if)# ip address 3.3.3.3 255.255.255.255
Delta(config)# interface loopback 1
Delta(config-if)# ip address 172.20.0.1 255.255.255.0

! — Beta (AS 100) —————
Beta(config)# interface loopback 0
Beta(config-if)# ip address 4.4.4.4 255.255.255.255
Beta(config)# interface loopback 1
Beta(config-if)# ip address 10.100.1.1 255.255.255.0
```

Step 2 – Configure eBGP Multihop Session

```
! — Delta (AS 147) —————
Delta(config)# router bgp 147
Delta(config-router)# bgp router-id 3.3.3.3
Delta(config-router)# no auto-summary
Delta(config-router)# no synchronization
Delta(config-router)# neighbor 4.4.4.4 remote-as 100
Delta(config-router)# neighbor 4.4.4.4 description eBGP-Multihop-to-Beta-Lo0
Delta(config-router)# neighbor 4.4.4.4 ebgp-multihop 2
Delta(config-router)# neighbor 4.4.4.4 update-source loopback 0
Delta(config-router)# network 172.20.0.0 mask 255.255.255.0
Delta(config-router)# end

! — Beta (AS 100) —————
Beta(config)# router bgp 100
Beta(config-router)# bgp router-id 4.4.4.4
Beta(config-router)# no auto-summary
Beta(config-router)# no synchronization
Beta(config-router)# neighbor 3.3.3.3 remote-as 147
Beta(config-router)# neighbor 3.3.3.3 description eBGP-Multihop-to-Delta-Lo0
```

```
Beta(config-router)# neighbor 3.3.3.3 ebgp-multihop 2
Beta(config-router)# neighbor 3.3.3.3 update-source loopback 0
Beta(config-router)# network 10.100.1.0 mask 255.255.255.0
Beta(config-router)# end
```

Step 3 – Verify Multihop Session

```
Beta# show bgp ipv4 unicast summary
Neighbor    V  AS    MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd
3.3.3.3      4  147    18        15        6     0     0    00:11:04  1

Beta# show bgp ipv4 unicast neighbors 3.3.3.3 | include hops
! Expected: 'External BGP neighbor may be up to 2 hops away'

Beta# show bgp ipv4 unicast 172.20.0.0/24
! Next-hop: 3.3.3.3, AS_PATH: 147

! Verify traceroute shows Echo in the path
Beta# traceroute 3.3.3.3 source 4.4.4.4
! Expected: 4.4.4.4 → 172.16.18.2 (Echo Gi0/1) → 3.3.3.3
```

5.3 Task 3 – iBGP: Apollo (Gi0/1) ↔ Beta (Gi0/0)

Apollo Gi0/1 (102.16.11.1) and Beta Gi0/0 (102.16.11.2) form the iBGP session within AS 100. next-hop-self is required on both sides so that each router replaces external eBGP next-hops with its own iBGP interface address, which the peer can resolve via OSPF.

```
! — Apollo (AS 100) —————
Apollo(config)# router bgp 100
Apollo(config-router)# neighbor 102.16.11.2 remote-as 100
Apollo(config-router)# neighbor 102.16.11.2 description iBGP-to-Beta-Gi0/0
Apollo(config-router)# neighbor 102.16.11.2 next-hop-self
Apollo(config-router)# end

! — Beta (AS 100) —————
Beta(config)# router bgp 100
Beta(config-router)# neighbor 102.16.11.1 remote-as 100
Beta(config-router)# neighbor 102.16.11.1 description iBGP-to-Apollo-Gi0/1
Beta(config-router)# neighbor 102.16.11.1 next-hop-self
Beta(config-router)# end

! Verify — Charlie's Lol should now appear on Beta via iBGP
Beta# show bgp ipv4 unicast 192.168.180.0/24
! next-hop should be 102.16.11.1 (Apollo), not 102.10.20.1 (Charlie)

! Verify Delta's Lol should now appear on Apollo via iBGP
Apollo# show bgp ipv4 unicast 172.20.0.0/24
! next-hop should be 102.16.11.2 (Beta), not 3.3.3.3 (Delta)
```

6. Part 3: Configure MP-BGP for IPv6

On IOS-XE, all IPv6 BGP neighbor commands are placed inside 'address-family ipv6 unicast'. There is no need to declare IPv6 neighbors under 'router bgp' first — doing so causes the '%Specify the peer-group first' error. The remote-as, description, ebgp-multihop, update-source, and next-hop-self commands all go inside the address-family block.

6.1 IPv6 Addressing Plan

Router	Interface	IPv6 Link/Loopback Address	Purpose
Charlie	Gi0/0	2001:db8:1::1/64	eBGP peering link with Apollo
Charlie	Lo1	2001:db8:180::1/64	Advertised IPv6 customer prefix
Apollo	Gi0/0	2001:db8:1::2/64	eBGP peering link with Charlie
Apollo	Gi0/1	2001:db8:a::1/64	iBGP peering link with Beta
Apollo	Lo1	2001:db8:100:1::1/64	Advertised IPv6 prefix for AS 100
Beta	Gi0/0	2001:db8:a::2/64	iBGP peering link with Apollo
Beta	Gi0/1	2001:db8:3::1/64	Transit link to Echo
Beta	Lo0	2001:db8:100::2/128	Multihop update-source (IPv6)
Beta	Lo1	2001:db8:100:2::1/64	Advertised IPv6 prefix for AS 100
Echo	Gi0/1	2001:db8:3::2/64	Transit link to Beta
Echo	Gi0/0	2001:db8:2::2/64	Link to Delta
Delta	Gi0/0	2001:db8:2::1/64	Link to Echo
Delta	Lo0	2001:db8:147::1/128	Multihop update-source (IPv6)
Delta	Lo1	2001:db8:147:1::1/64	Advertised IPv6 prefix for AS 147

6.2 Task 4 – Configure IPv6 Addresses on All Routers

```
! Enable on ALL routers first
Router(config)# ipv6 unicast-routing

! — Charlie —————
Charlie(config)# interface GigabitEthernet0/0
Charlie(config-if)# ipv6 address 2001:db8:1::1/64
Charlie(config-if)# ipv6 enable
Charlie(config)# interface loopback 1
Charlie(config-if)# ipv6 address 2001:db8:180::1/64

! — Apollo —————
Apollo(config)# interface GigabitEthernet0/0
Apollo(config-if)# ipv6 address 2001:db8:1::2/64
Apollo(config)# interface GigabitEthernet0/1
Apollo(config-if)# ipv6 address 2001:db8:a::1/64
Apollo(config)# interface loopback 1
Apollo(config-if)# ipv6 address 2001:db8:100:1::1/64
```

```

! — Beta —————
Beta(config)# interface GigabitEthernet0/0
Beta(config-if)# ipv6 address 2001:db8:a::2/64
Beta(config)# interface GigabitEthernet0/1
Beta(config-if)# ipv6 address 2001:db8:3::1/64
Beta(config)# interface loopback 0
Beta(config-if)# ipv6 address 2001:db8:100::2/128
Beta(config)# interface loopback 1
Beta(config-if)# ipv6 address 2001:db8:100:2::1/64

! — Echo —————
Echo(config)# interface GigabitEthernet0/1
Echo(config-if)# ipv6 address 2001:db8:3::2/64
Echo(config)# interface GigabitEthernet0/0
Echo(config-if)# ipv6 address 2001:db8:2::2/64

! — Delta —————
Delta(config)# interface GigabitEthernet0/0
Delta(config-if)# ipv6 address 2001:db8:2::1/64
Delta(config)# interface loopback 0
Delta(config-if)# ipv6 address 2001:db8:147::1/128
Delta(config)# interface loopback 1
Delta(config-if)# ipv6 address 2001:db8:147:1::1/64

```

6.3 Task 5 – Configure OSPFv3 (Delta, Echo, Beta)

OSPFv3 is the IPv6 equivalent of OSPFv2. It uses the same area structure but is configured per-interface rather than using the 'network' statement. OSPFv3 must be running between Delta, Echo, and Beta so that the IPv6 loopbacks (2001:db8:147::1/128 on Delta and 2001:db8:100::2/128 on Beta) are mutually reachable across Echo before the IPv6 BGP Multihop session is configured.

OSPFv3 Interface and Process Summary

Router	Interface	IPv6 Address	OSPFv3 Process	Area	Passive
Delta	Lo0	2001:db8:147::1/128	Process 1	Area 0	Yes
Delta	Lo1	2001:db8:147:1::1/64	Process 1	Area 0	Yes
Delta	Gi0/0	2001:db8:2::1/64	Process 1	Area 0	No
Echo	Gi0/0	2001:db8:2::2/64	Process 1 (AS147)	Area 0	No
Echo	Gi0/1	2001:db8:3::2/64	Process 1 (AS147) + Process 2 (AS100)	Area 0	No
Beta	Gi0/1	2001:db8:3::1/64	Process 2 (AS100 side)	Area 0	No
Beta	Lo0	2001:db8:100::2/128	Process 2	Area 0	Yes

KEY

On IOS-XE the correct OSPFv3 command is 'router ospfv3 PROCESS-ID' (not 'ipv6 router ospf'). Per-interface the command is 'ospfv3 PROCESS-ID ipv6 area AREA' (not 'ipv6 ospf PROCESS area'). Address families are configured inside the process block. This is the unified IOS-XE OSPFv3 syntax supporting both IPv4 and IPv6 under the same process.

Step 1 – Configure OSPFv3 on Delta (AS 147)

```
! — Delta (AS 147) —————
Delta(config)# router ospfv3 1
Delta(config-router)# router-id 3.3.3.3
Delta(config-router)# address-family ipv6 unicast
Delta(config-router-af)# passive-interface loopback 0
Delta(config-router-af)# passive-interface loopback 1
Delta(config-router-af)# exit-address-family
Delta(config-router)# exit

Delta(config)# interface loopback 0
Delta(config-if)# ospfv3 1 ipv6 area 0
Delta(config)# interface loopback 1
Delta(config-if)# ospfv3 1 ipv6 area 0
Delta(config)# interface GigabitEthernet0/0
Delta(config-if)# ospfv3 1 ipv6 area 0
```

Step 2 – Configure OSPFv3 on Echo (Transit — Two Processes)

Echo runs two OSPFv3 processes: process 1 for AS 147 (with Delta on Gi0/0) and process 2 for AS 100 (with Beta on Gi0/1). Gi0/1 is included in both processes so both domains can learn the path through Echo.

```
! — Echo —————
! AS 147 OSPFv3 process
Echo(config)# router ospfv3 1
Echo(config-router)# router-id 5.5.5.5
Echo(config-router)# address-family ipv6 unicast
Echo(config-router-af)# exit-address-family
Echo(config-router)# exit

Echo(config)# interface GigabitEthernet0/0
Echo(config-if)# ospfv3 1 ipv6 area 0      ! AS 147 side — link to Delta
Echo(config)# interface GigabitEthernet0/1
Echo(config-if)# ospfv3 1 ipv6 area 0      ! shared link — AS 147 sees Beta's
subnet

! AS 100 OSPFv3 process (separate process ID)
Echo(config)# router ospfv3 2
Echo(config-router)# router-id 5.5.5.5
Echo(config-router)# address-family ipv6 unicast
Echo(config-router-af)# exit-address-family
Echo(config-router)# exit

Echo(config)# interface GigabitEthernet0/1
Echo(config-if)# ospfv3 2 ipv6 area 0      ! AS 100 side — link to Beta
! Note: Gi0/1 is in BOTH process 1 and process 2 intentionally
```

Step 3 – Configure OSPFv3 on Beta (AS 100)

```
! — Beta (AS 100) —————
Beta(config)# router ospfv3 2
Beta(config-router)# router-id 4.4.4.4
Beta(config-router)# address-family ipv6 unicast
```

```

Beta(config-router-af)# passive-interface loopback 0
Beta(config-router-af)# exit-address-family
Beta(config-router)# exit

Beta(config)# interface loopback 0
Beta(config-if)# ospfv3 2 ipv6 area 0      ! Lo0 = 2001:db8:100::2/128 (multihop
source)
Beta(config)# interface GigabitEthernet0/1
Beta(config-if)# ospfv3 2 ipv6 area 0      ! link to Echo Gi0/1

```

Step 4 – Verify OSPFv3 Adjacencies and Loopback Reachability

```

! — Verify OSPFv3 neighbors —————
Delta# show ospfv3 neighbor
! Expected: Echo (5.5.5.5) on GigabitEthernet0/0 — FULL state

Beta# show ospfv3 neighbor
! Expected: Echo (5.5.5.5) on GigabitEthernet0/1 — FULL state

! — Verify IPv6 routes for loopbacks —————
Beta# show ipv6 route ospf
! Expected: O 2001:db8:147::1/128 via FE80::<Echo-link-local>, Gi0/1
! Expected: O 2001:db8:147:1::/64 via FE80::<Echo-link-local>, Gi0/1

Delta# show ipv6 route ospf
! Expected: O 2001:db8:100::2/128 via FE80::<Echo-link-local>, Gi0/0

! — Verify OSPFv3 process detail —————
Delta# show ospfv3 1 ipv6
! Shows process ID, router-id, area, and interface table
Beta# show ospfv3 2 ipv6

! — Critical: verify loopback pings BEFORE configuring BGP —
Delta# ping ipv6 2001:db8:100::2 source 2001:db8:147::1
Beta# ping ipv6 2001:db8:147::1 source 2001:db8:100::2
! Both must return !!!!! before proceeding to BGP configuration

! — Verify traceroute confirms Echo is in path —————
Beta# traceroute ipv6 2001:db8:147::1 source 2001:db8:100::2
! Expected: Lo0(Beta) → 2001:db8:3::2 (Echo Gi0/1) → 2001:db8:147::1 (Delta Lo0)

```

WARN

OSPFv3 uses link-local addresses (FE80::/10) for hello packets and next-hop in routing table entries. If the OSPFv3 adjacency is stuck in INIT or 2-WAY, verify that IPv6 is enabled on the interface ('ipv6 enable') and that link-local addresses are assigned.

6.4 Task 6 – MP-BGP IPv6: Charlie (AS 180) ↔ Apolo (AS 100)

All IPv6 neighbor commands go inside 'address-family ipv6 unicast' on IOS-XE. Two configuration options are shown for the Apolo and Beta iBGP session: Option A uses individual neighbor statements (simpler, suitable for small deployments), and Option B uses a peer-group (scalable — all shared attributes are defined once on the group and applied to all members).

Charlie and Apollo eBGP — individual neighbor statements

```
! — Charlie (AS 180) —————
Charlie(config)# router bgp 180
Charlie(config-router)# address-family ipv6 unicast
Charlie(config-router-af)# neighbor 2001:db8:1::2 remote-as 100
Charlie(config-router-af)# neighbor 2001:db8:1::2 description IPv6-eBGP-to-Apollo
Charlie(config-router-af)# neighbor 2001:db8:1::2 activate
Charlie(config-router-af)# network 2001:db8:180::/64
Charlie(config-router-af)# exit-address-family

! — Apollo (AS 100) — eBGP to Charlie —————
Apollo(config)# router bgp 100
Apollo(config-router)# address-family ipv6 unicast
Apollo(config-router-af)# neighbor 2001:db8:1::1 remote-as 180
Apollo(config-router-af)# neighbor 2001:db8:1::1 description IPv6-eBGP-to-Charlie
Apollo(config-router-af)# neighbor 2001:db8:1::1 activate
Apollo(config-router-af)# network 2001:db8:100:1::/64
Apollo(config-router-af)# exit-address-family
```

Option A — Apollo↔Beta iBGP: Individual Neighbor Statements

Straightforward — each neighbor command is applied directly to the specific peer. Suitable when there are few iBGP peers or when peers need different settings.

```
! — Apollo (AS 100) — Option A: individual iBGP to Beta —————
Apollo(config)# router bgp 100
Apollo(config-router)# address-family ipv6 unicast
Apollo(config-router-af)# neighbor 2001:db8:a::2 remote-as 100
Apollo(config-router-af)# neighbor 2001:db8:a::2 description IPv6-iBGP-to-Beta
Apollo(config-router-af)# neighbor 2001:db8:a::2 activate
Apollo(config-router-af)# neighbor 2001:db8:a::2 next-hop-self
Apollo(config-router-af)# exit-address-family

! — Beta (AS 100) — Option A: individual iBGP to Apollo —————
Beta(config)# router bgp 100
Beta(config-router)# address-family ipv6 unicast
Beta(config-router-af)# neighbor 2001:db8:a::1 remote-as 100
Beta(config-router-af)# neighbor 2001:db8:a::1 description IPv6-iBGP-to-Apollo
Beta(config-router-af)# neighbor 2001:db8:a::1 activate
Beta(config-router-af)# neighbor 2001:db8:a::1 next-hop-self
Beta(config-router-af)# network 2001:db8:100:2::/64
Beta(config-router-af)# exit-address-family
```

Option B — Apollo↔Beta iBGP: Peer-Group

A peer-group defines shared attributes once under a named group. Each peer is then assigned to the group, inheriting all its settings. This is the preferred approach in production when multiple iBGP peers share the same policy — adding a new iBGP speaker only requires one line (neighbor X peer-group GROUPNAME) instead of repeating every attribute. Choose Option A OR Option B — not both.

```
! — Apollo (AS 100) — Option B: peer-group for IPv6 iBGP —————
Apollo(config)# router bgp 100
Apollo(config-router)# address-family ipv6 unicast
```



```

Apolo(config-router-af)# ! Step 1: Define the peer-group and its shared
attributes
Apolo(config-router-af)# neighbor IPV6-IBGP-PEERS peer-group
Apolo(config-router-af)# neighbor IPV6-IBGP-PEERS remote-as 100
Apolo(config-router-af)# neighbor IPV6-IBGP-PEERS description IPv6-iBGP-peer-
group
Apolo(config-router-af)# neighbor IPV6-IBGP-PEERS next-hop-self
Apolo(config-router-af)# neighbor IPV6-IBGP-PEERS send-community
Apolo(config-router-af)# ! Step 2: Assign Beta to the peer-group
Apolo(config-router-af)# neighbor 2001:db8:a::2 peer-group IPV6-IBGP-PEERS
Apolo(config-router-af)# neighbor 2001:db8:a::2 activate
Apolo(config-router-af)# ! To add a future iBGP peer: just two lines
Apolo(config-router-af)# ! neighbor 2001:db8:a::3 peer-group IPV6-IBGP-PEERS
Apolo(config-router-af)# ! neighbor 2001:db8:a::3 activate
Apolo(config-router-af)# network 2001:db8:100:1::/64
Apolo(config-router-af)# exit-address-family

! — Beta (AS 100) — Option B: peer-group for IPv6 iBGP ———
Beta(config)# router bgp 100
Beta(config-router)# address-family ipv6 unicast
Beta(config-router-af)# neighbor IPV6-IBGP-PEERS peer-group
Beta(config-router-af)# neighbor IPV6-IBGP-PEERS remote-as 100
Beta(config-router-af)# neighbor IPV6-IBGP-PEERS description IPv6-iBGP-peer-group
Beta(config-router-af)# neighbor IPV6-IBGP-PEERS next-hop-self
Beta(config-router-af)# neighbor IPV6-IBGP-PEERS send-community
Beta(config-router-af)# neighbor 2001:db8:a::1 peer-group IPV6-IBGP-PEERS
Beta(config-router-af)# neighbor 2001:db8:a::1 activate
Beta(config-router-af)# network 2001:db8:100:2::/64
Beta(config-router-af)# exit-address-family

! — Verify peer-group ———
Apolo# show bgp ipv6 unicast peer-group IPV6-IBGP-PEERS
! Shows group attributes and all members
Apolo# show bgp ipv6 unicast summary
! Both Charlie (eBGP) and Beta (iBGP via peer-group) should show State/PfxRcd

```

TIP

Peer-group attributes act as defaults. A member can still override individual settings — e.g., 'neighbor 2001:db8:a::2 route-map POLICY in' applied directly to a member overrides the group for that attribute only. This gives flexibility without losing the scalability benefit.

6.5 Task 7 – MP-BGP IPv6 Multihop: Delta (AS 147) ↔ Beta (AS 100)

The IPv6 Multihop session uses Lo0 IPv6 addresses (2001:db8:147::1 on Delta and 2001:db8:100::2 on Beta) as update-source. OSPFv3 configured in Task 5 already advertises these loopbacks across Echo — no static routes needed.

```

! Verify IPv6 reachability first (OSPFv3 must be running)
Delta# ping ipv6 2001:db8:100::2 source 2001:db8:147::1
Beta# ping ipv6 2001:db8:147::1 source 2001:db8:100::2
! Both must succeed before configuring the BGP session

! — Delta (AS 147) ———
Delta(config)# router bgp 147
Delta(config-router)# address-family ipv6 unicast
Delta(config-router-af)# neighbor 2001:db8:100::2 remote-as 100
Delta(config-router-af)# neighbor 2001:db8:100::2 description IPv6-eBGP-Multihop-
to-Beta
Delta(config-router-af)# neighbor 2001:db8:100::2 ebgp-multihop 2

```

```

Delta(config-router-af)# neighbor 2001:db8:100::2 update-source loopback 0
Delta(config-router-af)# neighbor 2001:db8:100::2 activate
Delta(config-router-af)# network 2001:db8:147:1::/64
Delta(config-router-af)# exit-address-family

! — Beta (AS 100) —————
Beta(config)# router bgp 100
Beta(config-router)# address-family ipv6 unicast
Beta(config-router-af)# neighbor 2001:db8:147::1 remote-as 147
Beta(config-router-af)# neighbor 2001:db8:147::1 description IPv6-eBGP-Multihop-
to-Delta
Beta(config-router-af)# neighbor 2001:db8:147::1 ebgp-multihop 2
Beta(config-router-af)# neighbor 2001:db8:147::1 update-source loopback 0
Beta(config-router-af)# neighbor 2001:db8:147::1 activate
Beta(config-router-af)# exit-address-family

```

6.6 Task 8 – Verify All IPv6 Sessions

```

! — OSPFv3 check (must pass before checking BGP) —————
Delta# show ospfv3 neighbor
Beta# show ospfv3 neighbor
! Both should show Echo (5.5.5.5) in FULL state

! — On Apollo —————
Apollo# show bgp ipv6 unicast summary
Neighbor      V  AS      MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down
State/PfxRcd
2001:DB8:1::1 4  180     22       19       8       0    0    00:14:11  1
2001:DB8:A::2 4  100     18       16       8       0    0    00:11:30  1

! — On Beta —————
Beta# show bgp ipv6 unicast summary
Neighbor      V  AS      MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down
State/PfxRcd
2001:DB8:147::1 4  147     20       17       7       0    0    00:12:05  1
2001:DB8:A::1   4  100     18       16       8       0    0    00:11:30  1

! — If using peer-group (Option B) – verify group members ———
Apollo# show bgp ipv6 unicast peer-group IPV6-IBGP-PEERS
Beta# show bgp ipv6 unicast peer-group IPV6-IBGP-PEERS
! Shows group policy and all assigned members with their state

! — Verify prefix exchange —————
Apollo# show bgp ipv6 unicast 2001:db8:180::/64
Beta# show bgp ipv6 unicast 2001:db8:147:1::/64

! — End-to-end IPv6 reachability —————
Apollo# ping ipv6 2001:db8:180::1 source 2001:db8:100:1::1
! Charlie Lo1 → Apollo Lo1 – across eBGP direct session
Beta# ping ipv6 2001:db8:147:1::1 source 2001:db8:100:2::1
! Delta Lo1 → Beta Lo1 – across eBGP multihop session

```

WARN

If the IPv6 multihop session stays in Active: verify OSPFv3 is advertising Lo0 addresses, run the ping test sourced from Lo0, and confirm 'ebgp-multihop 2' and 'update-source loopback 0' are inside the address-family ipv6 unicast block on both Delta and Beta.

7. Part 4: BGP Best-Path Selection Exercises

7.1 Exercise A – Weight (Local to Beta)

Scenario: Beta should always prefer the eBGP Multihop path from Delta (3.3.3.3) for AS 147 prefixes over any iBGP-learned path from Apolo. Weight is local to the router and does not propagate.

```
Beta(config)# route-map PREFER-DELTA-EBGP permit 10
Beta(config-route-map)# set weight 300
Beta(config-route-map)# exit

Beta(config)# router bgp 100
Beta(config-router)# address-family ipv4 unicast
Beta(config-router-af)# neighbor 3.3.3.3 route-map PREFER-DELTA-EBGP in
Beta(config-router-af)# exit-address-family
Beta# clear ip bgp 3.3.3.3 soft in
Beta# show bgp ipv4 unicast 172.20.0.0/24
! Best path: weight 300, via 3.3.3.3
```

7.2 Exercise B – LOCAL_PREF (AS-wide)

Scenario: All routers in AS 100 should prefer exiting to AS 147 via Beta's multihop to Delta. Beta sets LOCAL_PREF=200 inbound from Delta; this propagates to Apolo via iBGP.

```
Beta(config)# route-map SET-LP-200 permit 10
Beta(config-route-map)# set local-preference 200
Beta(config-route-map)# exit

Beta(config)# router bgp 100
Beta(config-router)# address-family ipv4 unicast
Beta(config-router-af)# neighbor 3.3.3.3 route-map SET-LP-200 in
Beta(config-router-af)# exit-address-family
Beta# clear ip bgp 3.3.3.3 soft in
Apolo# show bgp ipv4 unicast 172.20.0.0/24
! localpref 200 should appear – received from Beta via iBGP
```

7.3 Exercise C – AS-Path Prepending

Scenario: Charlie prepends its own ASN twice outbound to Apolo, making its prefix appear less attractive if AS 100 had an alternate path.

```
Charlie(config)# route-map PREPEND permit 10
Charlie(config-route-map)# set as-path prepend 180 180
Charlie(config-route-map)# exit

Charlie(config)# router bgp 180
Charlie(config-router)# address-family ipv4 unicast
Charlie(config-router-af)# neighbor 102.10.20.2 route-map PREPEND out
Charlie(config-router-af)# exit-address-family
Charlie# clear ip bgp 102.10.20.2 soft out
```

```
Apolo# show bgp ipv4 unicast 192.168.180.0/24
! AS_PATH: 180 180 180 (original + 2 prepends)
```

7.4 Exercise D – MED

Scenario: AS 100 wants Delta to prefer entering AS 100 via Beta. Beta advertises AS 100 prefixes with a low MED to Delta.

```
Beta(config)# route-map SET-MED-50 permit 10
Beta(config-route-map)# set metric 50
Beta(config-route-map)# exit

Beta(config)# router bgp 100
Beta(config-router)# address-family ipv4 unicast
Beta(config-router-af)# neighbor 3.3.3.3 route-map SET-MED-50 out
Beta(config-router-af)# exit-address-family
Beta# clear ip bgp 3.3.3.3 soft out
Delta# show bgp ipv4 unicast 10.100.1.0/24
! metric 50 should appear in the path attributes
```

8. Part 5: BGP MD5 Authentication

8.1 Concepts

BGP MD5 authentication protects BGP sessions from spoofed TCP segments and unauthorised peering attempts. It uses a shared password configured identically on both peers. The password is hashed using MD5 into each TCP segment — if the hash does not match, the segment is silently dropped. On IOS-XE the command is 'neighbor X password PASSWORD'.

Key Point	Detail
Password must match exactly	Case-sensitive on both peers. A mismatch causes the session to drop silently.
Applied per-neighbor	Each BGP neighbor statement has its own password.
Session resets on apply	Adding or changing a password immediately resets the BGP session.
Verify mismatch	'show bgp ipv4 unicast neighbors X' shows 'MD5 authentication' and error counts.
IPv6 sessions	Same command applies inside address-family ipv6 unicast on IOS-XE.
Password encryption	Use 'service password-encryption' or type-7/type-6 in production.

WARN

A password mismatch is one of the most common BGP troubleshooting issues. The session shows 'Active' or 'Idle' with no error message. Always verify with 'debug ip bgp X.X.X.X events' or check TCP MD5 error counters in 'show bgp neighbors'.

8.2 Task 8 – Configure MD5 Authentication: Charlie ↔ Apolo (eBGP Direct)

Apply MD5 authentication on the direct eBGP session between Charlie and Apolo. Both routers must use exactly the same password.

```
! — Charlie (AS 180) —————
Charlie(config)# router bgp 180
Charlie(config-router)# neighbor 102.10.20.2 password BGP-LAB-180-100
Charlie(config-router)# end

! — Apolo (AS 100) —————
Apolo(config)# router bgp 100
Apolo(config-router)# neighbor 102.10.20.1 password BGP-LAB-180-100
Apolo(config-router)# end

! Session will briefly reset then re-establish with MD5
Apolo# show bgp ipv4 unicast neighbors 102.10.20.1 | include MD5|password
! Expected: 'MD5 authentication is enabled'
!           'Inbound soft reconfiguration allowed'

! Verify session re-established
Apolo# show bgp ipv4 unicast summary | include 102.10.20.1
! State/PfxRcd should show prefix count (not Active/Idle)
```

8.3 Task 9 – Configure MD5 Authentication: Delta ↔ Beta (eBGP Multihop)

Apply authentication on the multihop session. The password must match on both loopback-sourced BGP sessions. A different password from the direct session is used here to demonstrate per-session password management.

```
! — Delta (AS 147) —————
Delta(config)# router bgp 147
Delta(config-router)# neighbor 4.4.4.4 password BGP-LAB-147-100
Delta(config-router)# end

! — Beta (AS 100) —————
Beta(config)# router bgp 100
Beta(config-router)# neighbor 3.3.3.3 password BGP-LAB-147-100
Beta(config-router)# end

Beta# show bgp ipv4 unicast neighbors 3.3.3.3 | include MD5
! Expected: 'MD5 authentication is enabled'
```

8.4 Task 10 – Configure MD5 Authentication: Apolo ↔ Beta (iBGP)

iBGP sessions within AS 100 should also be authenticated to prevent rogue routers from forming iBGP sessions inside the AS.

```
! — Apolo (AS 100) —————
Apolo(config)# router bgp 100
Apolo(config-router)# neighbor 102.16.11.2 password BGP-LAB-IBGP
Apolo(config-router)# end

! — Beta (AS 100) —————
Beta(config)# router bgp 100
Beta(config-router)# neighbor 102.16.11.1 password BGP-LAB-IBGP
Beta(config-router)# end

! Verify all three sessions have MD5 enabled
Apolo# show bgp ipv4 unicast summary
! All neighbors should show prefix count — not Active or Idle

Apolo# show bgp ipv4 unicast neighbors 102.10.20.1 | include MD5
Apolo# show bgp ipv4 unicast neighbors 102.16.11.2 | include MD5
Beta# show bgp ipv4 unicast neighbors 3.3.3.3 | include MD5
```

8.5 Task 11 – MD5 Authentication for IPv6 Sessions (IOS-XE)

On IOS-XE, the password command for IPv6 neighbors goes inside 'address-family ipv6 unicast', consistent with how all other IPv6 neighbor commands are configured.

```
! — Charlie (AS 180) — IPv6 eBGP to Apolo —————
Charlie(config)# router bgp 180
Charlie(config-router)# address-family ipv6 unicast
Charlie(config-router-af)# neighbor 2001:db8:1::2 password BGP-LAB-V6-180-100
```

```

Charlie(config-router-af)# exit-address-family

! — Apollo (AS 100) — IPv6 eBGP to Charlie —————
Apollo(config)# router bgp 100
Apollo(config-router)# address-family ipv6 unicast
Apollo(config-router-af)# neighbor 2001:db8:1::1 password BGP-LAB-V6-180-100
Apollo(config-router-af)# exit-address-family

! — Apollo iBGP to Beta — IPv6 —————
Apollo(config)# router bgp 100
Apollo(config-router)# address-family ipv6 unicast
Apollo(config-router-af)# neighbor 2001:db8:a::2 password BGP-LAB-V6-IBGP
Apollo(config-router-af)# exit-address-family

Beta(config)# router bgp 100
Beta(config-router)# address-family ipv6 unicast
Beta(config-router-af)# neighbor 2001:db8:a::1 password BGP-LAB-V6-IBGP
Beta(config-router-af)# exit-address-family

! — Delta ↔ Beta IPv6 Multihop —————
Delta(config)# router bgp 147
Delta(config-router)# address-family ipv6 unicast
Delta(config-router-af)# neighbor 2001:db8:100::2 password BGP-LAB-V6-147-100
Delta(config-router-af)# exit-address-family

Beta(config)# router bgp 100
Beta(config-router)# address-family ipv6 unicast
Beta(config-router-af)# neighbor 2001:db8:147::1 password BGP-LAB-V6-147-100
Beta(config-router-af)# exit-address-family

! Verify IPv6 sessions with MD5
Apollo# show bgp ipv6 unicast neighbors 2001:db8:1::1 | include MD5
Beta# show bgp ipv6 unicast neighbors 2001:db8:147::1 | include MD5

```

8.6 Task 12 – Simulate and Verify Authentication Failure

Understanding what a password mismatch looks like is as important as configuring authentication correctly. This task deliberately introduces a mismatch then verifies the symptoms and restores the correct password.

```

! — Step 1: Introduce a mismatch on Apollo toward Charlie ———
Apollo(config)# router bgp 100
Apollo(config-router)# neighbor 102.10.20.1 password WRONG-PASSWORD
Apollo(config-router)# end

! — Step 2: Observe the symptoms —————
Apollo# show bgp ipv4 unicast summary | include 102.10.20.1
! State should transition to Active (session dropped)

Apollo# show bgp ipv4 unicast neighbors 102.10.20.1 | include MD5|TCP|error
! Look for: 'MD5 authentication is enabled'
! Look for: TCP connection statistics showing reset/refused packets

! Enable debug to see real-time mismatch events
Apollo# debug ip bgp 102.10.20.1 events
! Expected log: '%BGP-3-NOTIFICATION: sent to neighbor 102.10.20.1
!               passive open failed - TCB is not available'
! Or: 'MD5 hash failure from 102.10.20.1'

```



```
Apolo# undebug all

! — Step 3: Restore correct password —————
Apolo(config)# router bgp 100
Apolo(config-router)# neighbor 102.10.20.1 password BGP-LAB-180-100
Apolo(config-router)# end

! — Step 4: Verify session recovers —————
Apolo# show bgp ipv4 unicast summary | include 102.10.20.1
! State/PfxRcd should return to prefix count within 30-60 seconds
```

WARN

To remove MD5 authentication from a session, use 'no neighbor X password'. The session will reset and re-establish without authentication. Never change a password on a production session without a maintenance window.

9. Verification & Troubleshooting Reference

9.1 BGP Session Verification

Command	Purpose
show bgp ipv4 unicast summary	All IPv4 session states, prefix counts, uptime
show bgp ipv6 unicast summary	All IPv6 session states, prefix counts, uptime
show bgp ipv4 unicast neighbors X include MD5	Verify MD5 authentication status
show bgp ipv4 unicast neighbors X include hops	Verify ebgp-multihop TTL value
show bgp ipv4 unicast X.X.X.X/nn	Detailed path including next-hop, weight, LP, MED
show bgp ipv4 unicast neighbors X advertised-routes	What this router sends to the peer
show bgp ipv4 unicast neighbors X received-routes	What the peer sent (needs soft-reconfiguration)
show ip route bgp	BGP routes installed in the RIB
traceroute X source Y	Confirm number of hops and path through Echo
debug ip bgp X events	Real-time BGP events including MD5 failures

9.2 Common Issues & Solutions

Symptom	Likely Cause	Fix
Multihop session Active	OSPF not advertising loopbacks across Echo	Verify OSPF includes Lo0 and 172.16.18.0/24 on Beta and Echo
Multihop session Active	ebgp-multihop value too low	Confirm ebgp-multihop 2 on both Delta and Beta
Multihop session Active	update-source not configured	Add 'neighbor X update-source loopback 0' on both peers
Session Active after adding MD5	Password mismatch between peers	Check case-sensitivity; verify with debug ip bgp X events
Prefix in BGP table, not RIB	Next-hop unreachable (missing next-hop-self)	Add 'neighbor X next-hop-self' on iBGP sessions
No IPv6 prefixes received	Neighbor not declared inside address-family ipv6	All IPv6 neighbor commands must be inside address-family ipv6 unicast
OSPFv3 adjacency stuck in INIT	IPv6 not enabled on interface	Add 'ipv6 enable' to the interface; verify link-local address assigned
OSPFv3 Lo0 not in routing table	Loopback not added to OSPFv3 process	Add 'ospfv3 PROC ipv6 area 0' on the loopback interface — not 'ipv6 ospf'
Peer-group member not activating	activate missing for the member	Add 'neighbor X activate' for each member after assigning to peer-group
network statement not working	Prefix not in routing table	Verify 'show ip route X.X.X.X' shows an exact match (connected/static)

10. Review Questions

Answer all questions based on the confirmed topology. IPv4 and IPv6 configurations use the addressing tables in Sections 1.2 and 6.1 respectively.

10.1 Conceptual Questions

Q1. Why is `ebgp-multihop 2` the correct value for the Delta↔Beta session, and what security risk does setting it higher than necessary introduce?

The path from Delta Lo0 (3.3.3.3) to Beta Lo0 (4.4.4.4) is exactly 2 hops: Delta Gi0/0 → Echo → Beta Gi0/1. `ebgp-multihop N` sets the TTL of the BGP TCP packets to N. Setting it to exactly 2 ensures packets from a third router (3 hops away) are dropped.

Security risk: Setting a higher value (e.g., `ebgp-multihop 5`) means a router 3, 4, or 5 hops away could potentially send BGP packets that reach Delta or Beta. If an attacker can spoof the loopback IP and is within the TTL range, they could attempt to hijack or disrupt the BGP session. The principle of least privilege applies: always set `ebgp-multihop` to the exact hop count.

Q2. Why can the 172.16.18.0/24 link (Beta Gi0/1 ↔ Echo Gi0/1) be included in both AS 100 OSPF and AS 147 OSPF? Does this cause a routing loop?

Yes, it can be included in both OSPF domains — this is the same pattern as 102.10.20.0/24 being in AS 100 OSPF even though it connects to Charlie (AS 180). Including a link in an OSPF domain simply means that domain's routers know how to reach that subnet. It does not mean OSPF routes are exchanged between the two ASes.

No routing loop occurs because: (1) AS 100 OSPF and AS 147 OSPF are completely separate domains with no redistribution between them. (2) Beta only runs AS 100 OSPF; Delta only runs AS 147 OSPF. Echo runs both as separate processes. (3) BGP provides the inter-AS routing policy and loop prevention via `AS_PATH`.

Q3. Why does the `'network 192.168.180.0 mask 255.255.255.0'` statement on Charlie require that Lo1 be configured first?

The BGP `'network'` statement does NOT create a prefix — it only injects a prefix into BGP if an EXACT match already exists in the router's routing table (RIB). If Charlie's Lo1 (192.168.180.1/24) is not configured, the route 192.168.180.0/24 does not exist in the RIB, so BGP silently ignores the network statement and the prefix never appears in `'show bgp ipv4 unicast'`.

This is a common mistake: the network statement looks correct in `'show run'` but the prefix never gets advertised. Always verify with `'show ip route 192.168.180.0'` on Charlie first.

Q4. On IOS-XE, why must IPv6 BGP neighbors be declared inside `'address-family ipv6 unicast'` rather than under `'router bgp'`?

IOS-XE's BGP implementation treats the 'router bgp' level as the IPv4 default address family. When you attempt to declare an IPv6 address as a neighbor at that level, IOS-XE cannot determine which address family it belongs to without a peer-group context, triggering the '%Specify the peer-group first' error.

Inside 'address-family ipv6 unicast', the context is unambiguous — all neighbor commands (remote-as, activate, ebgp-multihop, update-source, next-hop-self, password) are applied within the IPv6 address family. This is the correct IOS-XE design and produces cleaner, more readable configuration than the IOS classic approach.

10.2 Configuration Questions

Q5. Write the complete IOS-XE configuration for the IPv6 eBGP Multihop session between Delta and Beta, including OSPFv3 neighbour statements and the address-family ipv6 unicast block. Use the addressing from Section 6.1.

```
! — OSPFv3 on Delta (ensures Lo0 IPv6 is reachable) —————
Delta(config)# router ospfv3 1
Delta(config-router)# router-id 3.3.3.3
Delta(config-router)# address-family ipv6 unicast
Delta(config-router-af)# passive-interface loopback 0
Delta(config-router-af)# exit-address-family
Delta(config)# interface loopback 0
Delta(config-if)# ospfv3 1 ipv6 area 0
Delta(config)# interface GigabitEthernet0/0
Delta(config-if)# ospfv3 1 ipv6 area 0

! — OSPFv3 on Echo (both processes) —————
Echo(config)# router ospfv3 1
Echo(config-router)# router-id 5.5.5.5
Echo(config-router)# address-family ipv6 unicast
Echo(config-router-af)# exit-address-family
Echo(config)# interface GigabitEthernet0/0
Echo(config-if)# ospfv3 1 ipv6 area 0
Echo(config)# interface GigabitEthernet0/1
Echo(config-if)# ospfv3 1 ipv6 area 0

Echo(config)# router ospfv3 2
Echo(config-router)# router-id 5.5.5.5
Echo(config-router)# address-family ipv6 unicast
Echo(config-router-af)# exit-address-family
Echo(config)# interface GigabitEthernet0/1
Echo(config-if)# ospfv3 2 ipv6 area 0

! — OSPFv3 on Beta —————
Beta(config)# router ospfv3 2
Beta(config-router)# router-id 4.4.4.4
Beta(config-router)# address-family ipv6 unicast
Beta(config-router-af)# passive-interface loopback 0
Beta(config-router-af)# exit-address-family
Beta(config)# interface loopback 0
Beta(config-if)# ospfv3 2 ipv6 area 0
Beta(config)# interface GigabitEthernet0/1
Beta(config-if)# ospfv3 2 ipv6 area 0

! Verify before BGP
```

```

Delta# ping ipv6 2001:db8:100::2 source 2001:db8:147::1

! — Delta (AS 147) — BGP —————
Delta(config)# router bgp 147
Delta(config-router)# address-family ipv6 unicast
Delta(config-router-af)# neighbor 2001:db8:100::2 remote-as 100
Delta(config-router-af)# neighbor 2001:db8:100::2 description IPv6-eBGP-Multihop-
to-Beta
Delta(config-router-af)# neighbor 2001:db8:100::2 ebgp-multihop 2
Delta(config-router-af)# neighbor 2001:db8:100::2 update-source loopback 0
Delta(config-router-af)# neighbor 2001:db8:100::2 activate
Delta(config-router-af)# network 2001:db8:147:1::/64
Delta(config-router-af)# exit-address-family

! — Beta (AS 100) — BGP —————
Beta(config)# router bgp 100
Beta(config-router)# address-family ipv6 unicast
Beta(config-router-af)# neighbor 2001:db8:147::1 remote-as 147
Beta(config-router-af)# neighbor 2001:db8:147::1 description IPv6-eBGP-Multihop-
to-Delta
Beta(config-router-af)# neighbor 2001:db8:147::1 ebgp-multihop 2
Beta(config-router-af)# neighbor 2001:db8:147::1 update-source loopback 0
Beta(config-router-af)# neighbor 2001:db8:147::1 activate
Beta(config-router-af)# exit-address-family

```

Q6. Configure MD5 authentication on the Delta↔Beta eBGP Multihop session and then simulate a mismatch. What command shows the authentication failure without enabling debug?

Configuration:

```

Delta(config-router)# neighbor 4.4.4.4 password MULTIROP-KEY
Beta(config-router)# neighbor 3.3.3.3 password MULTIROP-KEY

```

Simulate mismatch — change password on Beta only:

```

Beta(config-router)# neighbor 3.3.3.3 password WRONG-KEY

```

Show authentication failure without debug:

```

Beta# show bgp ipv4 unicast neighbors 3.3.3.3

```

Look for: 'MD5 authentication is enabled'

And: 'Connections established X; dropped Y' — Y increments on mismatch

And: 'Last reset X ago, due to BGP Notification sent'

```

Beta# show tcp brief

```

The TCP session will show CLOSED or SYN_SENT repeatedly (failing to establish)

Restore:

```

Beta(config-router)# neighbor 3.3.3.3 password MULTIROP-KEY

```

10.3 Scenario Questions

Q7. After adding MD5 authentication on Apolo toward Charlie, the session drops and stays in Active. Charlie's configuration was not changed yet. Describe exactly what is happening at the TCP level and how to resolve it.

What is happening: Apolo is now hashing its BGP TCP segments with MD5 using the new password. Charlie still expects unauthenticated TCP segments (no MD5 hash). When Charlie receives a TCP segment with an MD5 option it does not recognise, the OS-level TCP stack rejects the segment and sends a TCP RST. Apolo's BGP session transitions to Active as each connection attempt is reset.

Resolution: Configure the matching password on Charlie immediately:

```
Charlie(config)# router bgp 180
```

```
Charlie(config-router)# neighbor 102.10.20.2 password BGP-LAB-180-100
```

The session will re-establish within the BGP hold-timer interval (default up to 60 seconds for a new TCP connection attempt).

Best practice: In production, stage the configuration change during a maintenance window and configure both sides as quickly as possible to minimise outage duration.

Q8. On Beta, 'show bgp ipv4 unicast 192.168.180.0/24' shows the prefix with '>i' marker. What does 'i' mean here, what is the next-hop, and what would cause it NOT to be installed in the RIB?

'>i' means: '>' = best path selected; 'i' = learned via iBGP (as opposed to 'e' for eBGP or locally originated with no letter).

Next-hop: Because Apolo has 'next-hop-self' configured toward Beta, the next-hop is 102.16.11.1 (Apolo's iBGP interface address), NOT the original 102.10.20.1 (Charlie's Gi0/0).

Causes it would NOT be in the RIB (shows in BGP table but no '>'):

1. next-hop-self missing on Apolo — next-hop would be 102.10.20.1 which Beta cannot reach via OSPF.
2. BGP synchronization enabled — route must also be in the IGP.
3. OSPF not running or Apolo's 102.16.11.1 not reachable from Beta (unlikely if iBGP session is up, but possible if OSPF adjacency dropped).

Q9. You need to remove MD5 authentication from the Apolo↔Beta iBGP session without causing a permanent loss of BGP routes. Describe the safest procedure.

1. Ensure both routers have all routes already in their BGP tables (session is fully established and stable).
2. Remove the password on BOTH routers as quickly as possible — the session will reset on the first change:

```
Apolo(config-router)# no neighbor 102.16.11.2 password
```

```
Beta(config-router)# no neighbor 102.16.11.1 password
```
3. The iBGP session will drop immediately on the first 'no password' command. Routes learned via iBGP will be withdrawn from the BGP table on the other router.
4. BGP will re-establish the TCP session without MD5. The session should come back up within the hold-timer negotiation period (typically 30–60 seconds).

5. Verify: 'show bgp ipv4 unicast summary' — confirm session re-established and prefix counts match previous values.
6. Confirm: 'show bgp ipv4 unicast neighbors 102.16.11.2 | include MD5' — should show 'MD5 authentication is not enabled'.

In production: perform this during a maintenance window. If Apolo and Beta are the only iBGP speakers, all inter-AS routes will be temporarily unavailable during the session reset.

Q10. Challenge — Charlie advertises 192.168.180.0/24 (Lo1) to Apolo. Trace the complete path this prefix takes through the network: which BGP attributes are set at each hop, what next-hop changes occur, and where is MD5 authentication verified?

Step 1 — Charlie → Apolo (eBGP direct, MD5 authenticated with BGP-LAB-180-100):

UPDATE sent: NLRI=192.168.180.0/24, NEXT_HOP=102.10.20.1 (Charlie Gi0/0), AS_PATH=180, ORIGIN=IGP.

MD5 hash verified by Apolo on the TCP segment.

Apolo installs the route: next-hop=102.10.20.1, reachable via OSPF (102.10.20.0/24 in AS 100 OSPF).

Step 2 — Apolo → Beta (iBGP, MD5 authenticated with BGP-LAB-IBGP):

UPDATE sent: NLRI=192.168.180.0/24, NEXT_HOP=102.16.11.1 (due to next-hop-self), AS_PATH=180, LOCAL_PREF=100 (default), ORIGIN=IGP.

MD5 hash verified by Beta on the TCP segment.

Beta installs the route: next-hop=102.16.11.1, reachable via OSPF. Route shows as '>i' in Beta's BGP table.

Step 3 — Beta → Delta (eBGP Multihop, MD5 authenticated with BGP-LAB-147-100):

If Beta re-advertises to Delta: NEXT_HOP=4.4.4.4 (Beta Lo0, eBGP sets NH to self), AS_PATH=100 180.

LOCAL_PREF is stripped (iBGP-only attribute, not sent to eBGP peers).

MD5 hash verified by Delta on the multihop TCP segment (2 hops via Echo).

Delta installs: next-hop=4.4.4.4, reachable via OSPF (4.4.4.4/32 in AS 147 OSPF via Echo).

11. Lab Completion Checklist

Verify each item with the appropriate show command before signing off.

#	Objective	Status
1	OSPF: Delta Lo0 (3.3.3.3) reachable from Beta and Apolo via OSPF — no static routes	☐
2	OSPF: Beta Lo0 (4.4.4.4) reachable from Delta via OSPF through Echo	☐
3	IPv4 eBGP direct: Charlie Gi0/0 ↔ Apolo Gi0/0 — Established, 192.168.180.0/24 received	☐
4	IPv4 eBGP Multihop: Delta Lo0 ↔ Beta Lo0 — Established, ebgp-multihop 2 confirmed	☐
5	IPv4 iBGP: Apolo Gi0/1 ↔ Beta Gi0/0 — Established, next-hop-self verified	☐
6	Charlie's Lo1 (192.168.180.0/24) visible on Beta via iBGP with next-hop 102.16.11.1	☐
7	OSPFv3: Delta Lo0 (2001:db8:147::1) reachable from Beta — ping from 2001:db8:100::2	☐
8	OSPFv3: Beta Lo0 (2001:db8:100::2) reachable from Delta — ping from 2001:db8:147::1	☐
9	OSPFv3: Delta, Echo, Beta adjacencies all in FULL state (show ospfv3 neighbor)	☐
10	MP-BGP IPv6: Charlie ↔ Apolo — Established, 2001:db8:180::/64 received on Apolo	☐
11	MP-BGP IPv6 iBGP: Apolo ↔ Beta — configured via Option A (individual) OR Option B (peer-group)	☐
12	If Option B: show bgp ipv6 unicast peer-group IPV6-IBGP-PEERS confirms members and policy	☐
13	MP-BGP IPv6 Multihop: Delta Lo0 ↔ Beta Lo0 — Established, ebgp-multihop 2 in AF block	☐
14	Exercise A: Weight 300 on Beta inbound from Delta verified	☐
15	Exercise B: LOCAL_PREF 200 from Beta propagated to Apolo via iBGP	☐
16	Exercise C: AS-PATH 180 180 180 on Charlie's prefix confirmed on Apolo	☐
17	Exercise D: MED 50 on AS 100 prefixes confirmed on Delta	☐
18	MD5 Auth: Charlie↔Apolo — 'MD5 authentication is enabled' confirmed	☐
19	MD5 Auth: Delta↔Beta — 'MD5 authentication is enabled' confirmed	☐
20	MD5 Auth: Apolo↔Beta iBGP — 'MD5 authentication is enabled' confirmed	☐
21	MD5 Auth: All IPv6 sessions authenticated — verified with show bgp ipv6	☐
22	MD5 mismatch simulated and session recovery verified	☐
23	End-to-end IPv4: ping 192.168.180.1 from Delta (via BGP path)	☐

#	Objective	Status
24	End-to-end IPv6: ping 2001:db8:180::1 from 2001:db8:147:1::1	☐

WARN	Save all configurations with 'write memory' on every router including Echo. Echo's OSPF configuration is critical for the multihop session and is easily forgotten since Echo runs no BGP.
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